



BAULKHAM HILLS HIGH SCHOOL

2015
YEAR 12
TERM 2 ASSESSMENTS

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work

Total marks – 45

- Attempt Questions 1 – 4
- All questions are NOT of equal value

Total marks – 45

Attempt Questions 1 – 4

All questions are NOT of equal value

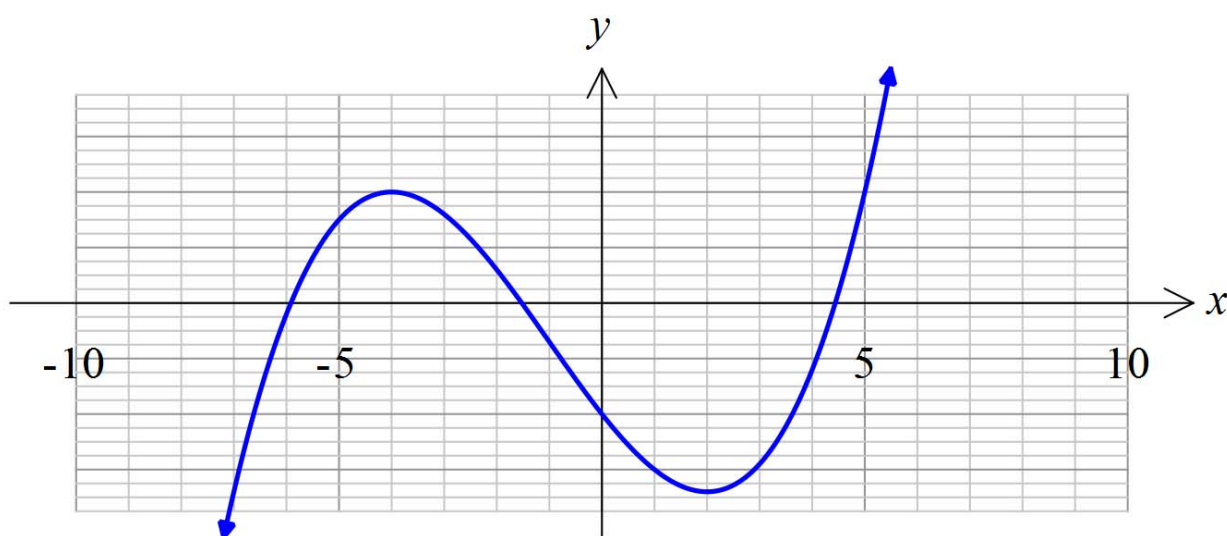
Answer each question on the appropriate answer sheet. Each answer sheet must show your BOS#. Extra paper is available.

Your responses should include relevant mathematical reasoning and/or calculations.

	Marks
Question 1 (12 marks) Use a <i>separate</i> piece of paper	
a) (i) Between which two integers does $\sqrt{5}$ lie?	1
(ii) Using repeated applications of the “halving the interval” method, to approximate $\sqrt{5}$, correct to one decimal place.	2
b) Use the given substitution to evaluate the following integrals.	
(i) $\int 6x\sqrt{9-x^2} \, dx$ using $u = 9-x^2$	3
(ii) $\int_0^{\frac{3}{2}} \frac{dx}{\sqrt{9-x^2}}$ using $x = 3\sin\theta$	3
(iii) $\int \frac{-\sin 2x}{2+3\cos^2 x} \, dx$ using $u = 2+3\cos^2 x$	3

Question 2 (9 marks) Use a *separate* piece of paper

a) Below is a graph of $y = x^3 + 3x^2 - 24x - 40$.



- (i) Explain why $x = 0$ would not be a good first approximation in order to find the positive solution to $x^3 + 3x^2 - 24x - 40 = 0$ using Newton's Method. 1
 - (ii) Give an example of an approximation to $x^3 + 3x^2 - 24x - 40 = 0$ that would fail to find any solution using Newton's Method, and explain why it would fail. 2
 - (iii) Using Newton's Method, find the positive solution to $x^3 + 3x^2 - 24x - 40 = 0$, correct to two decimal places. 3
- b) If $\ddot{x} = e^{-x}$ and initially the particle is observed to be at $x = 0$ with a velocity of 2 m/s, find v^2 as a function of x . 3

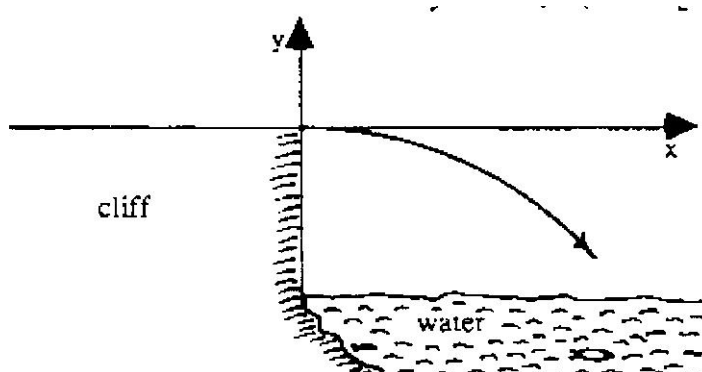
Question 3 (12 marks) Use a *separate* piece of paper

- a) The velocity of a particle moving along the x -axis is given by $v^2 = 36 - 6x - 2x^2$, where x is in metres.

(i) Prove that the particle is moving in Simple Harmonic Motion. 2

(ii) Find the path that the particle travels. 2

- b) An object is projected horizontally from the top of a vertical cliff 40 metres above sea level with a velocity of 40 m/s. (Take $g = 10 \text{ m/s}^2$)



- (i) Using the top edge of the cliff as the origin, prove that the equations of motion are given by; 3

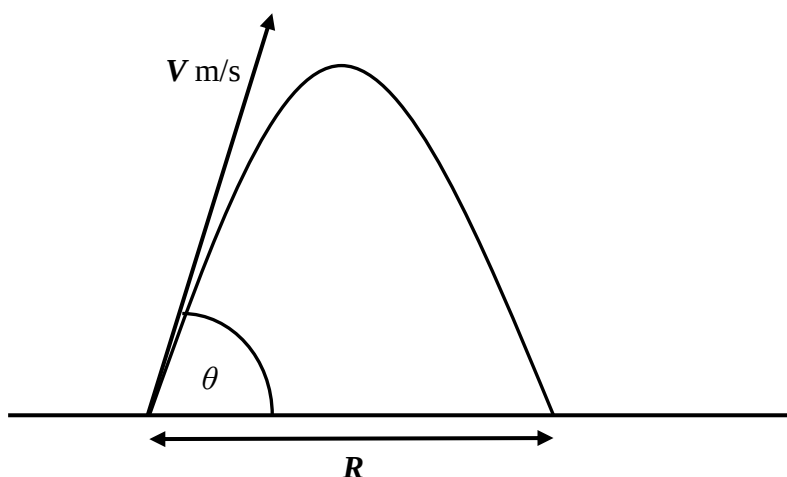
$$x = 40t \quad \text{and} \quad y = -5t^2$$

(ii) Calculate when and where the object hits the water. 2

(iii) Find the speed of the object and the angle it makes with the water, the instant it hits the water. 3

Question 4 (12 marks) Use a *separate* piece of paper

- a) A particle is travelling in simple harmonic motion such that its displacement x metres from the origin is given by $\ddot{x} = -4x$.
- (i) Show that $x = A\cos(2t + \beta)$ is a possible equation of motion for the particle, where A and β are positive constants. 2
- (ii) The particle is initially observed to have a velocity of 2 m/s and a displacement from the origin of 4 metres. Show that the amplitude of the motion is $\sqrt{17}$ metres. 2
- (iii) Determine the maximum speed of the particle. 1
- b) Sudarshan is in a pool holding a hose at an angle of θ to the surface of the water and water is leaving the hose with a velocity of V m/s. The water from the hose hits the surface of the water at a distance R metres from Sudarshan.



Using the equations of motion;

$$x = Vt\cos\theta \quad \text{and} \quad y = Vt\sin\theta - \frac{1}{2}gt^2 \quad (\text{Do NOT prove this})$$

- (i) Show that $R = \frac{V^2 \sin 2\theta}{g}$ 2
- (ii) Explain why the maximum range occurs when $\theta = 45^\circ$ 1
- (iii) Sudarshan moves the hose so that the angle θ changes at a constant rate $\frac{d\theta}{dt} = k$, from an initial value of $\theta = 30^\circ$. 2
- Prove that $\frac{dR}{dt} = \frac{2kV^2 \cos 2\theta}{g}$
- (iv) Find $\frac{d^2R}{dt^2}$ and hence show that the motion is simple harmonic. 2

~ END OF EXAMINATION ~

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

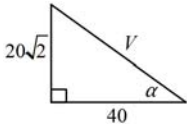
$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log x, \quad x > 0$

BAULKHAM HILLS HIGH SCHOOL
YEAR 12 EXTENSION 1 TERM 2 ASSESSMENT 2015 SOLUTIONS

Solution	Marks	Comments
QUESTION 1		
1 (a) (i) $\sqrt{5}$ lies in between 2 and 3	1	1 mark • Correct answer
1 (a) (ii) $f(x) = x^2 - 5 = 0$ $f(2) = 2^2 - 5 = -1 < 0$ $f(3) = 3^2 - 5 = 4 > 0$ $x_1 = \frac{2+3}{2} = 2.5$ $x_2 = \frac{2+2.5}{2} = 2.25$ $x_3 = \frac{2+2.25}{2} = 2.13$ $f(2.5) = (2.5)^2 - 5 = 1.25 > 0$ $f(2.25) = (2.25)^2 - 5 = 0.0625 > 0$ $f(2.13) = (2.13)^2 - 5 = -0.44631 < 0$ $\therefore 2 < \sqrt{5} < 2.5$ $\therefore 2 < \sqrt{5} < 2.25$ $\therefore 2.13 < \sqrt{5} < 2.25$ $x_4 = \frac{2.13 + 2.25}{2} = 2.19$ $f(2.19) = (2.19)^2 - 5 = -0.2039 < 0$ $\therefore 2.19 < \sqrt{5} < 2.25$ $\therefore \sqrt{5} = 2.2 \text{ correct to one decimal place}$	2	2 marks • Correct solution using an appropriate number of iterations 1 mark • Correctly applies the "halving the interval" method at least once
1 (b) (i) $\int 6x\sqrt{9-x^2} dx$ $= -3 \int \sqrt{u} du$ $= -3 \times \frac{2}{3} u\sqrt{u} + c$ $= -2(9-x^2)\sqrt{9-x^2} + c$ $u = 9 - x^2$ $du = -2x dx$	3	3 marks • Correct solution using the given substitution 2 marks • Correct primitive in terms of u 1 mark • Correct integrand in terms of u • Correctly finds answer using an alternative approach
1 (b) (ii) $\int_0^{\frac{3}{2}} \frac{dx}{\sqrt{9-x^2}} = \int_0^{\frac{\pi}{6}} \frac{3\cos\theta d\theta}{\sqrt{9-9\sin^2\theta}}$ $x = 3\sin\theta$ $dx = 3\cos\theta d\theta$ when $x = 0$, $\theta = 0$ when $x = \frac{3}{2}$, $\theta = \frac{\pi}{6}$ $= \int_0^{\frac{\pi}{6}} \frac{3\cos\theta d\theta}{3\cos\theta}$ $= \int_0^{\frac{\pi}{6}} d\theta$ $= [\theta]_0^{\frac{\pi}{6}}$ $= \frac{\pi}{6}$	3	3 marks • Correct solution using the given substitution • <i>Note: solving as an indefinite integral, then using answer to find definite integral is acceptable</i> 2 marks • Correct primitive in terms of θ • Correct integrand in terms of θ, including the correct limits 1 mark • Correct integrand in terms of θ without the limits • Correctly finds answer using an alternative approach
1 (b) (iii) $\int \frac{-\sin 2x}{2+3\cos^2 x} dx$ $= \frac{1}{3} \int \frac{du}{u}$ $= \frac{1}{3} \ln u + c$ $= \frac{1}{3} \ln(2+3\cos^2 x) + c$ $u = 2 + 3\cos^2 x$ $du = -6\cos x \sin x dx$ $= -3\sin 2x dx$	3	3 marks • Correct solution using the given substitution 2 marks • Correct primitive in terms of u 1 mark • Correct integrand in terms of u • Correctly finds answer using an alternative approach

Solution		Marks	Comments
QUESTION 2			
2(a) (i)	The tangent at $x = 0$ would cut the x-axis close to $x = -2$, which would be a good approximation to one of the negative solutions, but not the positive solution. This occurred because $x = 0$ is on the opposite side of the stationary point to the positive solution.	1	1 mark Valid explanation
2(a) (ii)	The approximation would fail at either stationary point, as the tangent would be horizontal meaning it will never cut the x-axis. In addition, if it is a stationary point then the derivative is zero at this point so zero would be substituted into the denominator of Newton's formula, thus making it undefined. Stationary points are $x = 2$ and $x = -4$	2	2 marks - Locate a correct example with a valid explanation 1 mark - Locates a correct example
2 (a) (iii)	$f(x) = x^3 + 3x^2 - 24x - 40 \quad f'(x) = 3x^2 + 6x - 24$ $x_0 = 4 \quad x_1 = 4 - \frac{f(4)}{f'(4)} \quad x_2 = 4.5 - \frac{f(4.5)}{f'(4.5)} \quad x_3 = 4.44 - \frac{f(4.44)}{f'(4.44)}$ $= 4 - \frac{-24}{48} = 4.5 - \frac{3.875}{63.75} = 4.44 - \frac{0.109184}{61.7808}$ $= 4.5 \quad = 4.44 \quad = 4.44$ $\therefore x = 4.44$ is the positive solution, correct to two decimal places	3	3 marks - Correctly finds the approximation to the positive solution 2 marks - Finds one of the negative solutions - Correctly uses Newton's Method at least once 1 mark - Attempts to apply Newton's Method by using a correct formula
2 (b)	$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = e^{-x}$ $\frac{1}{2}v^2 = -e^{-x} + c$ $v^2 = -2e^{-x} + c$ when $x = 0, v = 2$ $4 = -2e^0 + c$ $c = 6$ $v^2 = 6 - 2e^{-x}$ OR $v \frac{dv}{dx} = e^{-x}$ $\int_2^v v dv = \int_0^x e^{-x} dx$ $\left[\frac{1}{2}v^2\right]_2^v = \left[-e^{-x}\right]_0^x$ $\frac{1}{2}v^2 - 2 = -e^{-x} + 1$ $\frac{1}{2}v^2 = -e^{-x} + 3$ $v^2 = 6 - 2e^{-x}$	3	3 marks - Correct solution 2 marks - Finds an expression for v^2 using a correct method 1 mark - Identifies that $\ddot{x} = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$ or equivalent expression linking velocity and displacement

Solution		Marks	Comments
QUESTION 3			
3 (a) (i)	$v^2 = 36 - 6x - 2x^2$ $\frac{1}{2}v^2 = 72 - 12x - 4x^2$ $\ddot{x} = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$ $= -12 - 8x$ $= -8\left(x + \frac{3}{2}\right)$ $\therefore \ddot{x} = -n^2X, \text{ where } n = 2\sqrt{2} \text{ and } X = x + \frac{3}{2}$	2	2 marks • Correct solution 1 mark • Identifies the condition for SHM • Finds an expression for acceleration in terms of displacement
3 (a) (ii)	$v^2 \geq 0$ $36 - 6x - 2x^2 \geq 0$ $x^2 + 3x - 18 \leq 0$ $(x + 6)(x - 3) \leq 0$ $-6 \leq x \leq 3$ The particle oscillates between $x = -6$ and $x = 3$	2	2 marks • Correct solution 1 mark • Progress towards solution using valid methods
3 (b) (i)	$\ddot{x} = 0$ $\dot{x} = c_1$ when $t = 0, \dot{x} = 40$ $\therefore 40 = c_1$ $\dot{x} = 40$ $x = 40t + c_2$ when $t = 0, x = 0$ $\therefore 0 = 0 + c_2$ $c_2 = 0$ $x = 40t$ $\ddot{y} = -10$ $\dot{y} = -10t + c_3$ when $t = 0, \dot{y} = 0$ $\therefore 0 = 0 + c_3$ $c_3 = 0$ $\dot{y} = -10t$ $y = -5t^2 + c_4$ when $t = 0, y = 0$ $\therefore 0 = 0 + c_4$ $c_4 = 0$ $y = -5t^2$	3	3 marks • Correct solution 2 marks • Correctly proves equation of motion for either x or y • Finds both equations without explicitly finding the value of the constants. 1 mark • Finds equations for both $\dot{x}$ and $\dot{y}$
3 (b) (ii)	Object hits the water when $y = -40$ i.e. $-5t^2 = -40$ $t^2 = 8$ $t = 2\sqrt{2}$ Object hits the water after $2\sqrt{2}$ seconds, $80\sqrt{2}$ metres from the base of the cliff. When $t = 2\sqrt{2}$; $x = 40(2\sqrt{2})$ $= 80\sqrt{2}$	2	2 marks • Correct solution 1 mark • Finds when it hits the water • Finds where it hits the water.
3 (b) (iii)	When $t = 2\sqrt{2}$; $\dot{x} = 40 \quad \text{and} \quad \dot{y} = -10(2\sqrt{2}) = -20\sqrt{2}$  $V^2 = 40^2 + (20\sqrt{2})^2$ $= 2400$ $V = \sqrt{2400}$ $= 20\sqrt{6}$ $\tan \alpha = \frac{20\sqrt{2}}{40}$ $= \frac{1}{\sqrt{2}}$ $\alpha = 35.264\dots$ $\therefore$ object hits the water at a speed of $20\sqrt{6}$ m/s at an angle of 35° to the water	3	3 marks • Correct solution <i>Note: angle can be either acute or obtuse</i> 2 marks • Finds the velocity of the object • Finds the angle the object makes with the water (either acute or obtuse) 1 mark • Calculates the horizontal and vertical components of the velocity <i>Note: correct answers based upon time found in part (ii) should be marked correct.</i>

Solution	Marks	Comments
QUESTION 4		
4 (a) (i) $x = A\cos(2t + \beta)$ $\dot{x} = -2A\sin(2t + \beta)$ $\ddot{x} = -4A\cos(2t + \beta)$ $= -4x$ Thus $x = A\cos(2t + \beta)$ is a possible equation of motion.	2	2 marks • Correct solution 1 mark • Attempts to find acceleration as a function of time
4 (a) (ii) when $t = 0$, $x = 4$, $v = 2$; $x = A\cos(2t + \beta)$ $\dot{x} = -2A\sin(2t + \beta)$ $4 = A\cos\beta$ $2 = 2A\sin\beta$ $16 = A^2\cos^2\beta$ $4 = 4A^2\sin^2\beta$ $1 = A^2\sin^2\beta$ $A^2\cos^2\beta + A^2\sin^2\beta = 17$ $A^2 = 17$ $A = \sqrt{17}$ $\therefore$ amplitude of the motion is $\sqrt{17}$ metres	2	2 marks • Successfully shows result 1 mark • Uses initial conditions in a valid attempt to show the given result
4 (a) (iii) $\dot{x} = -2A\sin(2t + \beta)$ $\therefore$ maximum speed of the particle is $2A = 2\sqrt{17}$ m/s	1	1 mark • Correct answer
4 (b) (i) $x = Vt\cos\theta$ $y = Vt\sin\theta - \frac{1}{gt^2}$ when $x = R$; $R = Vt\cos\theta$ when $y = 0$; $t = \frac{R}{V\cos\theta}$ $0 = V\left(\frac{R}{V\cos\theta}\right)\sin\theta - \frac{1}{2}g\left(\frac{R}{V\cos\theta}\right)^2$ $0 = \frac{R\sin\theta}{\cos\theta} - \frac{gR^2}{2V^2\cos^2\theta}$ $0 = 2V^2R\sin\theta\cos\theta - gR^2$ $0 = R(2V^2\sin\theta\cos\theta - gR)$ $R = 0$ or $R = \frac{2V^2\sin\theta\cos\theta}{g}$ $= \frac{V^2\sin 2\theta}{g}$ But $R \neq 0$, $\therefore R = \frac{V^2\sin 2\theta}{g}$	2	2 marks • Correct solution 1 mark • Eliminates t from the parametric equations
4 (b) (ii) $\frac{V^2}{g}$ is constant, thus R is a maximum when $\sin 2\theta$ is a maximum maximum $\sin 2\theta$ occurs when $2\theta = 90^\circ$ i.e. $\theta = 45^\circ$	1	1 mark • Correct explanation
4 (b) (iii) $\frac{dR}{dt} = \frac{dR}{d\theta} \times \frac{d\theta}{dt}$ $= \frac{2V^2\cos 2\theta}{g} \times k$ $= \frac{2kV^2\cos 2\theta}{g}$	2	2 marks • Correct solution 1 mark • Finds $\frac{dR}{d\theta}$
4 (b) (iv) $\frac{d^2R}{dt^2} = \frac{d}{dt}\left(\frac{dR}{dt}\right)$ $= \frac{d}{d\theta}\left(\frac{dR}{dt}\right) \times \frac{d\theta}{dt}$ $= -\frac{4kV^2\sin 2\theta}{g} \times k$ $= -\frac{4k^2V^2\sin 2\theta}{g}$ $= -4k^2R$ $\therefore$ motion is SHM as $\ddot{x} = -n^2x$, where $n = 2k$	2	2 marks • Correct solution 1 mark • Finds $\frac{d^2R}{dt^2}$ in terms of θ